**2** **(a)** Define specific heat capacity.

.....

.....

..... [2]

**(b)** A fixed mass of water in a beaker is at atmospheric pressure.

**(i)** The initial temperature of the water is 0 °C.

The water is supplied with thermal energy $E$, so that its temperature increases to 8 °C. There is no net change in the volume of the water.

Use the first law of thermodynamics to complete Table 2.1 for this process.

**Table 2.1**

| work done on water | thermal energy supplied to water | increase in internal energy of water |
|---|---|---|
| | $+ E$ | |

[2]

**(ii)** The water is now heated so that its temperature increases by a further 8 °C to a final temperature of 16 °C. This process causes the volume of the water to increase so that work $W$ is done.
Assume that the change in internal energy is the same as in **(b)(i)**.

Use the first law of thermodynamics to complete Table 2.2 for this process.

**Table 2.2**

| work done on water | thermal energy supplied to water | increase in internal energy of water |
|---|---|---|
| | | |

[2]

**(c)** Use the information in **(b)** to suggest, with a reason, how the average specific heat capacity of water between 8 °C and 16 °C compares with its average value between 0 °C and 8 °C.

.....

..... [1]

[Total: 7]

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